



A multi-modal smart chest patch for real-time cardiopulmonary monitoring and anomaly detection

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ABSTRACT Cardiopulmonary homeostasis is critical for health, with disruptions often indicating early-stage pathologies. However, current wearable devices, constrained by their limited focus on single or few modalities, fall short in capturing the full spectrum of these critical interactions. In this work, we present a multimodal smart chest patch (SCP) system integrating flexible sensing modules with a multi-criteria, multimodal fusion (MCMF) machine learning model. The patch (5.4 g and 3.6 mm) simultaneously monitors electrocardiogram (ECG), heart sound (HS), and respiratory (Resp) signals, enabling real-time extraction of 12 cardiopulmonary parameters. Compared with commercial devices, our flexible patch maintains stable signal quality and demonstrates broad applicability across diverse individuals. The proposed MCMF model was further validated on an extended dataset of 5,561 recordings from 475 participants and achieved a classification accuracy of 87% for detecting cardiac and respiratory anomalies, surpassing conventional methods. Our results enable real-time exercise monitoring, revealing dynamic physiological shifts ($\Delta\text{HR} = 21$ bpm, $\Delta\text{PEP} = -30$ ms) with superior signal fidelity, and therefore holding significant promise for scalable, personalized health management.

Keywords: cardiopulmonary monitoring, smart chest patch, multimodal, anomaly detection, flexible electronics

INTRODUCTION

The cardiovascular and respiratory systems are intricately intertwined, forming a dynamic network that sustains physiological homeostasis through coordinated autonomic [1], biomechanical [2,3], and neurohumoral interactions [4]. This interdependence is vital, as the heart and lungs work synergistically to maintain life [5], with cardiac output driving oxygenated blood circulation and pulmonary gas exchange ensuring systemic oxygenation, all regulated by feedback loops controlling heart rate, respiratory rhythm, and vascular tone. Disruptions to this balance—such as cardiogenic pulmonary edema causing alveolar fluid accumulation [6] or chronic obstructive pulmonary disease (COPD) leading to right ventricular overload via hypoxic vasoconstriction [7]—highlight the critical need for comprehensive monitoring to detect pathological changes early.

Importantly, with cardiovascular and respiratory diseases accounting for 40% of global mortality [8], monitoring indicators like respiratory sinus arrhythmia (RSA), pre-ejection period (PEP), and heart rate variability (HRV) simultaneously is essential [9,10], as they reflect heart-lung interactions and provide insights into autonomic dysfunction and cardiopulmonary stress, enabling timely clinical intervention.

However, current wearable technologies are inadequate for capturing the complex dynamics of heart-lung interactions [11–27]. Single-modality devices, such as wrist-based pulse oximeters [12,13,23] or neck-worn acoustic sensors [14,15], measure isolated metrics like heart rate or oxygen saturation but are positioned far from the signal source, often yielding low signal-to-noise ratios (SNR). Moreover, multimodal wearable systems that synchronously acquire electrocardiogram (ECG), heart sound (HS), and respiratory (Resp) signals are rare [16,21,22,28], constrained by challenges in achieving millisecond-scale flexible sensor synchronization [17–19], integrating heterogeneous signal modalities [20,25,27], and detecting anomalies amidst physiological variability [11,24,26]. These limitations hinder the wearables to capture composite metrics, such as cardiopulmonary coupling indicators, and obscure subtle anomalies of the cardiopulmonary system. Critically, the scarcity of multi-dimensional cardiopulmonary datasets further limits the development of machine learning models capable of identifying complex pathological signatures.

To address these challenges, we present a light-weight smart chest patch (SCP, with 5.4 g weight and 3.6 mm thickness) system that integrates synchronous acquisition of ECG, HS, and Resp signals on a flexible substrate, via a custom-designed flexible circuit, enabling real-time extraction of 12 cardiopulmonary parameters, including RSA, PEP, HRV, respiratory rate (RR), and other cardiopulmonary related indicators. Positioned at the precordial site, our patch leverages direct proximity to the heart and lungs to achieve superior signal fidelity and resilience to interference (Fig. 1a), surpassing limitations of peripheral sensors. To validate the broad applicability of the SCP system, we conducted 3-modality data synchronization collection and screening of 12 cardiopulmonary parameters on subjects with different physical conditions. Additionally, to integrate heterogeneous signal modalities and identify cardiopulmonary dysfunction under different data combinations of these 3-

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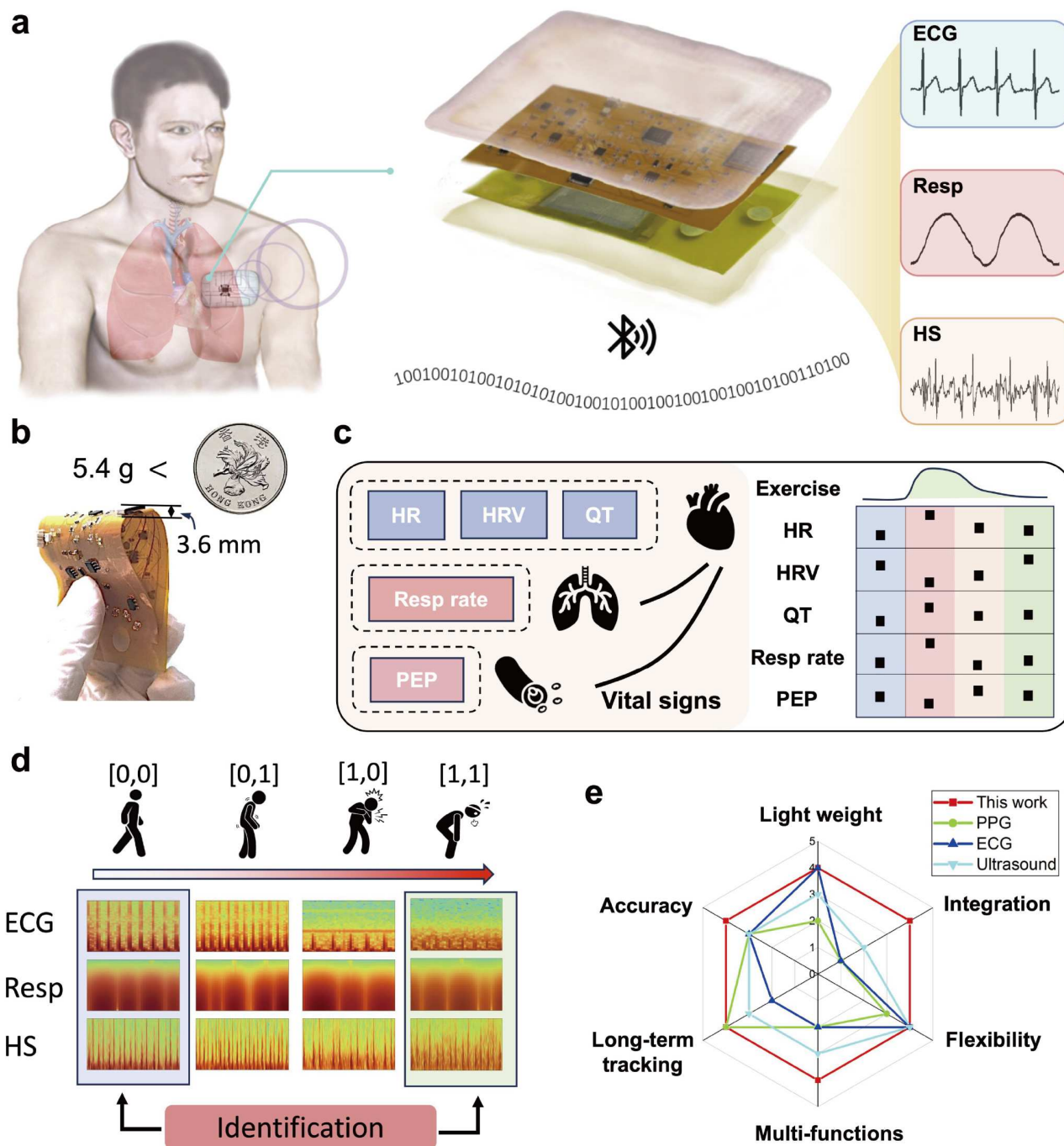


Figure 1 Conceptual diagram of the flexible smart chest patch (SCP) system. (a) Illustration of the SCP for cardiopulmonary function assessment, including flexible hardware, integrated HS, ECG, and respiratory sensors and signals, and wireless data transmission. (b) Excellent mechanical flexibility and an ultralight form factor of the SCP system. (c) The represented cardiopulmonary parameters that extracted from the ECG, HS, and Resp signals. (d) The identification of the cardiopulmonary dysfunction of the subjects, where the normal is labelled by [0,0], the respiratory dysfunction is labelled as [0,1], cardiac dysfunction is labelled as [1,0], and the cardiopulmonary dysfunction is labelled as [1,1]. (e) The technical comparison between our SCP system with photoplethysmography (PPG, green line), ultrasound (turquoise line), and ECG (blue line) sensors for continuous cardiopulmonary metrics in terms of multi-functions, long-term tracking, accuracy, light weight, integration, and flexibility.

modality signals, we developed a multi-criteria, multi-modal fusion (MCMF) machine learning model to achieve highly sensitive detection of cardiopulmonary abnormalities (the area under the receiver operating characteristic curve, i.e., $AUC > 0.92$). Beyond diagnostics, the SCP supports personalized exer-

cise management by providing real-time feedback on cardiopulmonary interaction metrics and risk evaluation. As a prototype, our SCP captures millisecond-level multi-modal signals to comprehensively monitor cardiopulmonary homeostasis, significantly enhancing wearable health monitoring capabilities.

EXPERIMENTAL SECTION

Fabrication of the sensing layer of the SCP system

The sensing layer of the SCP system was fabricated on a 25 μm -thick polyimide (PI) thin film. Two electrode arrays with uniform and serpentine geometries, designed for the I-lead ECG sensor and the respiratory sensor respectively, were patterned onto the PI substrate via laser processing and etching techniques. Subsequently, a HS sensor was fabricated using a multi-layer stacking method. The sensor consisted of five layers, arranged from top to bottom as follows: a top electrode layer, a grid-patterned polyurethane (PU) spacer, a polyvinylidene fluoride (PVDF) layer, another PU spacer, and a bottom electrode layer. Each adjacent layer was bonded using 50 μm -thick double-sided adhesive tape. The final dimensions of the HS sensor were 30 mm $\times$ 30 mm. Finally, a 30 mm $\times$ 30 mm window was cut out from the PI substrate via laser processing, and the HS sensor was integrated into the window using adhesive tape, forming the complete sensing layer of the SCP system with an overall size of 120 mm $\times$ 60 mm.

The integration of the SCP system

The SCP system was assembled by integrating the sensing layer with a flexible electronic circuit layer. A low-power micro-controller (32-bit) was employed for multi-channel signal acquisition from all three sensors, and a Bluetooth module (ESP32) was used for wireless data transmission to a personal computer. All electronic components, including capacitors, amplifiers, resistors, filters, and the analog-to-digital converter, were integrated onto a flexible printed circuit board (FPCB). The entire system was powered by a 5 V, 100 mAh lithium-ion battery. The final SCP device weighed approximately 5.4 g. Data acquisition was performed at a sampling rate of 500 Hz. Signal processing programs were developed using MATLAB. To reduce interference from movement artifacts, we applied specialized noise reduction filters with tailored bandpass cutoff frequencies for different signal modalities. To ensure biocompatibility and environmental protection, the entire system was encapsulated by a thin polydimethylsiloxane (PDMS) film during measurements.

Study population

We recruited three young males with varied body compositions—two with normal body fat percentages and one overweight—to evaluate our sensor system's cardiopulmonary parameter measurements, approved by the Joint Chinese University of Hong Kong–New Territories East Cluster Clinical Research Ethics Committee with written informed consent from all participants (Joint CUHK-NTEC CREC Ref. No.: 2022.335). They sat in a normal posture, wearing our device to capture three signals, compared with commercial devices measuring at 1000 Hz with millisecond-level resolution. One participant underwent exercise monitoring, including respiratory regulation training, to capture real-time, millisecond-level cardiovascular signal changes. Besides, the study utilized clinical datasets including: EPHNOGRAM [29,30], with 24 healthy adults (aged 23–29, average 25.4 $\pm$ 1.9 years) in 30-min stress-test sessions (resting, walking, running, biking); SensSmartTech [31], with 338 30-s recordings from 32 healthy volunteers across 10 channels (4 ECG, 1 HS, 4 PPG, 1 accelerometer(ACC)); PhysioNet/Computing in Cardiology Challenge 2016 [32], with 117 normal and 292 abnormal heart sound recordings; and MIMIC-I [33], with 10 pulmonary

edema and respiratory failure recordings.

Data collection

During data collection, participants sat relaxed in a chair or wore a chest patch during rest and exercise (running at specified speeds). Synchronized HS, ECG, and respiratory signals were recorded continuously using the commercial Biopac MP150 system (BIOPAC Systems Inc., California, USA) with ECG100C, PCG100C, and RSP100C modules, alongside our device. For dynamic signal measurement, the subject wore the SCP system throughout the experiment. The protocol consisted of a resting phase, followed by 5 min of elliptical exercise, and then a 20-min post-exercise resting phase. Data were collected at four time points: at rest (sitting), at the 5th min of exercise, after 5 min of rest (standing), and after 20 min of rest (sitting). We derived twelve cardiorespiratory parameters: heart rate (HR, i.e., 60/RR interval, RRI), HRV, QT interval ($QT_c = QT/\sqrt{RRI}$), S1 interval (S1 inv), S2 interval (S1 inv), S1/S2 ratio, RR, inspiratory duration (TI), expiratory duration (TE), PEP (R-S1), and Sys duration (R-S2), RSA, calculated as $\max(RRI) - \min(RRI)$ within a respiratory cycle. Parameters were extracted using time-domain methods (e.g., peak detection for HR and RR), variability analysis (e.g., RMSE for HRV), and time-frequency analysis (e.g., short time power spectrum estimate).

Model development

A machine learning pipeline has been developed to assess the risk of cardiac and respiratory diseases using ECG, HS, and respiratory data collected by a 3-modal flexible device in static resting state and dynamic states of resting, exercise, post-exercise, and recovery. The three modalities of signals were subjected to short-time Fourier transform (STFT) to extract sufficient time and frequency analysis. Features from all modalities were standardized, dimension-reduced via principal component analysis (PCA) [34] (retaining first 6, 12, and 18 principal components), and then concatenated across different modalities. 10 classifiers (support vector machine, *k*-nearest neighbors, logistic regression, linear discriminant analysis, random forest, decision tree, AdaBoost, gradient boosting, bagged trees, and naive Bayes) were optimized using strict regularization and class weight parameter adjustments. Five-fold cross-validation based on 85% of the training set guided the technique for MCMF model (detailed step-by-step process see Supplementary information Note S1), and performance metrics (accuracy, AUC, specificity, sensitivity) [35] were evaluated on the 15% test set. Single-modality and state-specificity analyses confirmed the superiority and robustness of the multimodal model.

Statistical analysis

Model performance was evaluated using sensitivity, specificity, accuracy, and AUC. Parameter reliability was assessed via Pearson correlation with reference measurements. Group differences were analyzed using paired *t*-tests, with a significance threshold of $P < 0.05$. Statistical analyses were performed in MATLAB (R2024b).

RESULTS AND DISCUSSION

Working principle of the SCP system

The SCP system presented herein represents an advanced, fully integrated, and wearable hemodynamic monitoring platform,

designed for continuous, multimodal cardiopulmonary assessment. It integrates the synchronous acquisition of ECG, HS, and respiratory signals, enabling comprehensive and multi-dimensional evaluation of cardiovascular and respiratory function through a single on-body device. As shown in Fig. 1a, the SCP system is conformally mounted on the chest, where its sensing region is positioned over the aortic valve—a location that has been validated as optimal for acquiring high-fidelity HS signals [36]. The system comprises three vertically integrated layers: a sensing layer, a flexible circuit layer, and a PDMS encapsulation layer. This multilayered architecture supports multimodal signal acquisition, embedded signal processing, and bluetooth low energy (BLE)-based wireless data transmission, while ensuring robust signal coordination and minimal latency (Fig. 2). Fig. 1b shows that the SCP system exhibits excellent mechanical flexibility and an ultralight form factor, weighing only 5.4 g, which is even lighter than a Hong Kong one-dollar coin. This lightweight design minimizes interference with natural body movement and ensures wearer comfort during prolonged usage.

The integrated design enables simultaneous extraction of 12 cardiorespiratory parameters, including cardiac-related, respiratory-related indicators, and the cardiopulmonary coupling indicators. Among them, five representative parameters HR, HRV, RR, QT, RSA, and PEP are highlighted in Fig. 1c to illustrate the SCP's multimodal monitoring capability. Specifically, cardiac function is captured through high-fidelity ECG, which records electrical activity, and HS, which detects mechanical heart sounds (S1, S2, and pathological S1/S2 ratio), while respiratory dynamics are quantified via chest wall motion, yielding parameters such as RR and inspiratory-to-expiratory (I:E) ratio. The interdependence of these systems—governed by autonomic neural regulation and biomechanical coupling—is further elucidated through advanced cross-modal analyses, including ECG-HS coherence and ECG-respiration phase synchronization. These cardiorespiratory parameters enable precise characterization of heart-lung interactions, which is critical for detecting subtle anomalies in conditions such as cardiac diseases or respiratory failure and pulmonary edema. The detailed methodology for extracting all 12 parameters will be presented later.

Beyond the heterogeneous signals and parameters extraction, later we will derive a multimodal fusion model that leverages spectral and rhythmic features across ECG, HS, and respiratory signals, and achieves robust discrimination of cardiorespiratory states across diverse physiological conditions (Fig. 1d). A radar chart in Fig. 1e benchmarks the SCP system against widely used sensing technologies showing that our system outperforms these modalities in terms of flexibility, signal diversity, long-term tracking capability, integration level, and overall wearability, which highlights its potential as a novel, medical-grade health monitoring solution.

Fabrication, circuit design, and signal validation of the SCP system

The integrated SCP system leverages a system-on-chip (SoC) with BLE capabilities for data acquisition, time synchronization, and coordinated data transmission. The fabrication process is illustrated in Fig. 2a, where the triboelectric-based HS sensor is fabricated using a layer-by-layer method. A grid-patterned PU spacer is first prepared via near-field electrospinning to serve as

a flexible structural framework. Silver nanowires (AgNWs) are then spray-coated onto the PU layers to form the conductive electrodes. A PVDF film is deposited by far-field electrospinning to act as the negative triboelectric layer. The PU spacer exhibits a well-defined grid geometry, while the PVDF layer forms a porous fibrous network, as confirmed by the SEM image (Fig. S1). Afterwards, an additional AgNW electrode and PU grid spacer are added to the opposite side of the PVDF layer, resulting in a bilaterally symmetric configuration. These materials are laminated into a multilayer triboelectric nanogenerator (TENG) structure, which enables sensitive detection of mechanical vibrations associated with heartbeats. Detailed information on the HS sensor fabrication can be found in our previous work [36]. Later, this HS sensor is integrated into a flexible PI substrate alongside other sensing components. The respiratory sensor, which is based on a strain gauge mechanism, features a serpentine conductive pattern designed to enhance strain sensitivity and mechanical compliance. The ECG electrodes are patterned around the HS sensor in order to ensure optimal chest placement and consistent skin contact. All ECG and respiratory sensors are co-fabricated onto the same PI substrate via laser processing and etching techniques, which ensures compactness, mechanical stability, and reduced motion impacts. Fig. S2a and b jointly validate the rationality of the SCP system's sensing layer design, where the respiratory sensor exhibits reliable responses to different levels of breathing, and the stable long-term ECG recordings confirm consistent signal acquisition from the chest-mounted configuration. In addition, we performed motion-based stability validation of the SCP while subjects were paddling in Fig. S3, showing good signal-to-noise ratios compared with commercial Biopac systems. We acquired signals from a subject at 48-h intervals for long-term stability assessment, see Fig. S4.

The complete SCP system is shown in Fig. 2b, which illustrates both the multimodal sensing layer and the flexible circuit layer. As shown in Fig. S5, the flexible circuit layer adopts a single-layer architecture in order to enhance compactness, while integrating key functional modules, including low-noise current amplifiers, filters, high-resolution analog-to-digital converters (ADCs), and a Bluetooth module for wireless data transmission (Fig. S6). A system schematic in Fig. 2c outlines the signal acquisition and processing pipeline, from multimodal sensing to wireless transmission and data visualization on personal computers. The final assembled device measures 12 cm × 6 cm, offering a lightweight and portable solution for continuous cardiopulmonary tracking with 500 Hz sampling rate (i.e., 2 ms temporal resolution). As demonstrated in Fig. 2d, the signals obtained from the SCP system across ECG, HS, and respiratory sensors are compared with those recorded using a commercial Biopac system. The high consistency between the two validates the signal fidelity, robustness, and clinical relevance of the SCP system, thereby confirming its capability for accurate, real-time, and wearable cardiopulmonary monitoring.

Extraction of cardiopulmonary parameters via the SCP system

To validate the robustness and adaptability of our SCP system, we conducted cardiopulmonary assessments across a diverse cohort of subjects using the device. Fig. 3a, b illustrate the characteristics of the three signal modalities and their corresponding states, which can be readily applied to the human chest region for real-time, non-invasive monitoring of cardio-

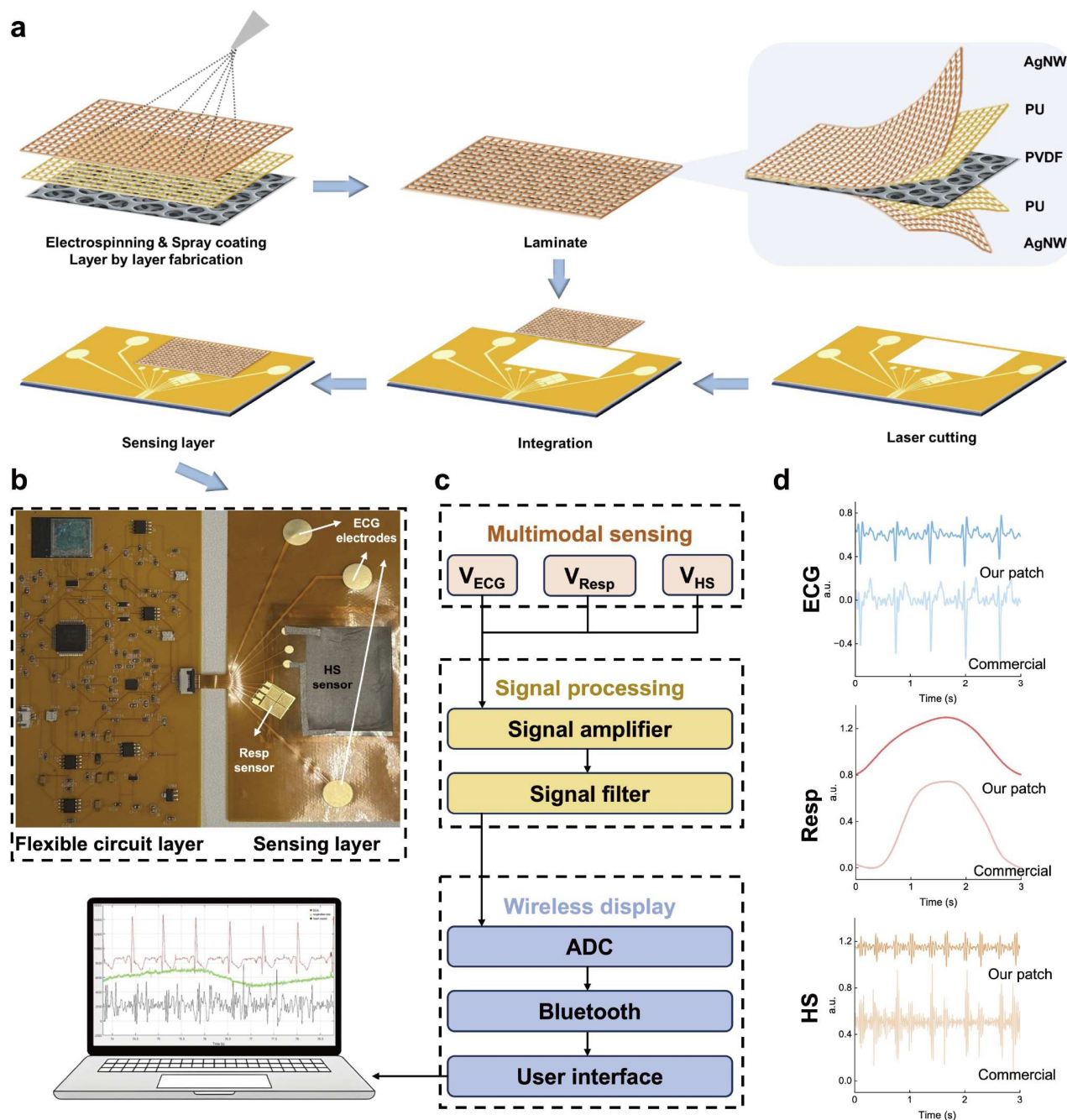


Figure 2 Fabrication, circuit design, and signal validation of the SCP system. (a) Fabrication process of the SCP system. (b) Photographs of the sensing layer and the flexible circuit layer of the SCP system. (c) Workflow of the SCP system. (d) Synchronized ECG, Resp, and HS signals from the SCP system compared with commercial devices.

pulmonary function (Movie S1). These signals concurrently yield 12 cardiopulmonary parameters: HRV, RRI, QTc, S1 inv, S2 inv, S1/S2 ratio, RR, TI, TE, RSA, PEP, and Sys duration, etc. Notably, the interactive metrics from the heart and lungs, such as RSA and PEP, are associated with cardiac and pulmonary pathologies [37].

To illustrate clinical applicability, synchronized measurements from three representative participants—two with similar body fat percentages (P1 and P3) and one with obesity (P2)—revealed distinct physiological profiles (Fig. 3d). The obese participant

exhibited elevated HR (>90 bpm) and RR (>20 bpm), alongside reduced HRV and RSA, reflecting autonomic dysregulation and cardiopulmonary strain. Comparative analysis of the 12 parameters across participants highlighted amplified cardiopulmonary coupling deficits in obesity, with statistically significant differences in QTc and PEP ($P < 0.01$, paired t -tests). These findings underscore the SCP system's sensitivity to physiological variations, enabling robust monitoring across diverse populations. Notably, unlike conventional wearable sensors, which are limited to isolated metrics (e.g., HR, RR) and per-

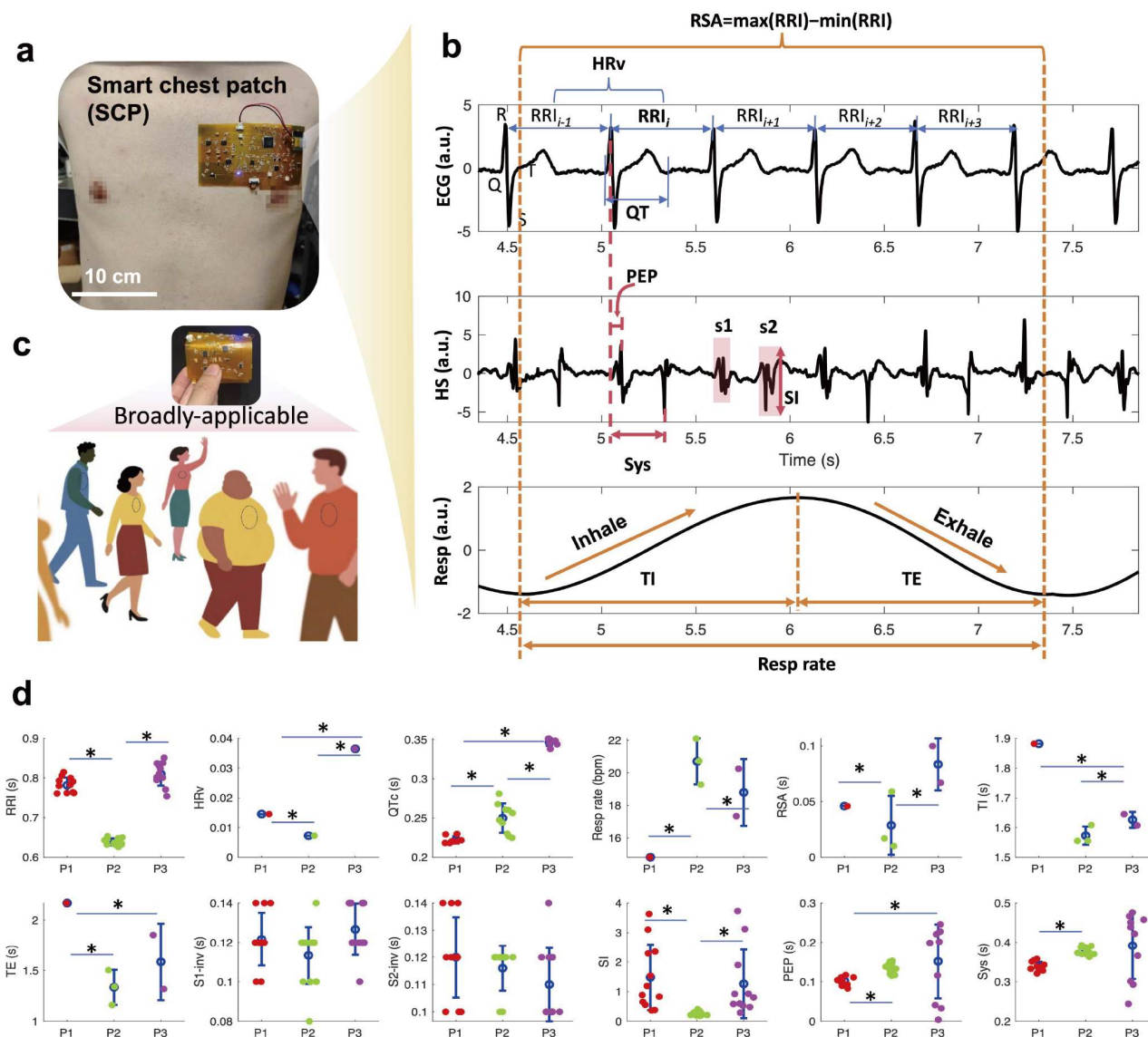


Figure 3 Extraction of cardiopulmonary parameters via the SCP system. (a) SCP attached to the subject's chest. (b) Typical 3 signal modalities, including ECG, HS, and respiratory signals, and the corresponding 12 cardiopulmonary parameters. (c) Cartoon illustration depicting the demographic versatility of the SCP system. (d) 12 cardiopulmonary parameters extracted by the SCP system among 3 subjects with diverse body characteristics.

ipheral placements [38,39], our precordial SCP system integrates multimodal signals to deliver a comprehensive cardiorespiratory assessment.

Multimodal cardiorespiratory assessment model

To harness the synergistic potential of multimodal signals for precision cardiorespiratory monitoring, we developed an MCMF health assessment framework that integrates the ECG, HS, and respiratory signals for robust and accurate physiological state classification. The MCMF model architecture, illustrated in Fig. 4a, seamlessly fuses both automatically spectral and rhythmic features acquired from open source datasets (EPHNOGRAM [29,30], SensSmartTech [31], PhysioNet/Computing in Cardiology Challenge 2016 [32], and MIMIC-I [33]), comprising 5,561 recordings from 475 participants categorized as normal, cardiac disease, and respiratory failure in Fig. 4b. These recordings were divided into training, validation, and test

datasets (Fig. 4c). Central to our framework is a multimodal fusion model that accounts for inter-modality variations and their synergistic effects on cardiorespiratory function (Fig. 4d). We employed PCA with unidirectional projection [34] to embed high-dimensional multimodal features into a low-dimensional subspace, preserving critical temporal and spectral information while reducing computational complexity. A dual-channel classification architecture was designed to process ECG-HS and ECG-respiration correlations independently, capturing cardiac and pulmonary dynamics with high fidelity. These channels feed into a weighted fusion classifier (Fig. 4e, f), optimized through iterative training and evaluation metrics to mitigate overfitting and enhance generalizability. The classifier categorizes cardiopulmonary status into discrete categories, such as normal, cardiac abnormality, and pulmonary abnormality, and outputs a comprehensive cardiopulmonary dysfunction score (CDS), which is the cardiac and respiratory risk probability generated by

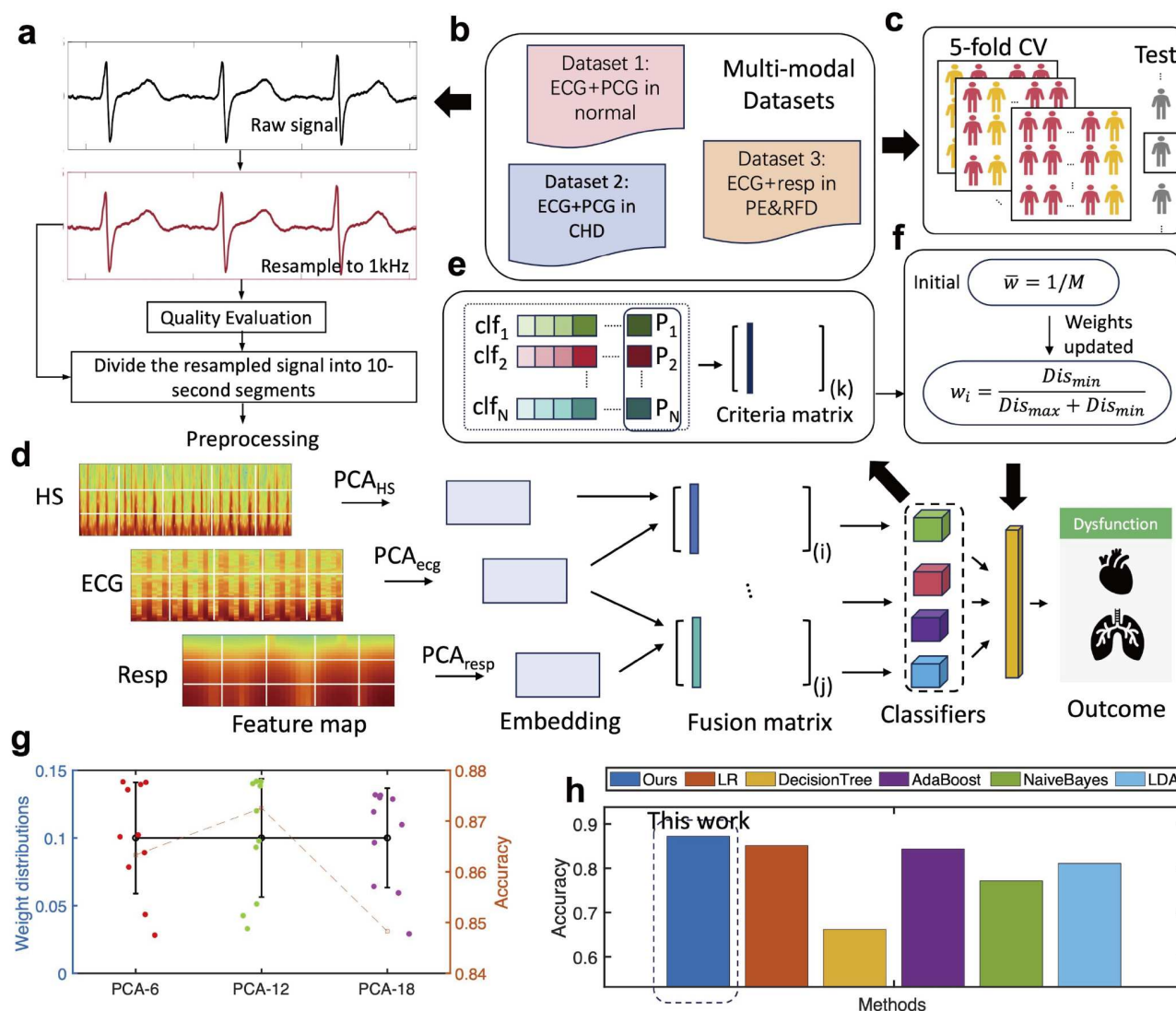


Figure 4 Modelling and assessment of the multimodal cardiopulmonary fusion model (MCMF). (a–f) Model architecture including (a) signal processing, (b) multimodal datasets, including normal dataset, patients with cardiac disease (CHD), and patients with pulmonary edema and respiratory failure diseases (PE&RFD), (c) comprehensive evaluation of the datasets under 5-fold cross-validation and testing dataset, (d) the spectrums of 3 modalities signal under PCA compression [34] and fusion process, (e) multi-criteria evaluation module, and (f) updating weights for multi-classifiers. (g) The weight distribution of the classifiers under different PCA compression. (h) The accuracy comparison of the proposed multimodal and multi-criteria fusion model and the existing ML methods.

the classifier.

We systematically evaluated the impact of PCA dimensions (6, 12, and 18 components, respectively) on model accuracy and generalization ability (Fig. 4g). Optimal performance was achieved with 12 components, balancing feature retention and noise suppression, with higher dimensions introducing extraneous variability that diminished classification accuracy. Comparative analyses against conventional machine learning approaches (e.g., decision tree, AdaBoost) demonstrate our MCMF model's superior performance, attributable to the weighted fusion strategy and multimodal integration (Fig. 4h). This framework outperforms existing methods, achieving a classification accuracy of 87% (with AUC > 0.92) and robustness across diverse physiological conditions. By leveraging synchronized multimodal signals, our approach provides a transformative paradigm for early detection and classification of

cardiorespiratory anomalies, offering profound implications for precision diagnostics and personalized health management.

Real-time cardiopulmonary monitoring for exercise optimization

To demonstrate the potential of our SCP system as a comprehensive tool for monitoring how the cardiopulmonary system responds to exercise, we further developed an advanced computational framework that integrates ECG, HS, and respiratory signals. This framework enables dynamic tracking of physiological states during different exercise phases—pre-exercise (rest), exercise, post-exercise, and recovery (Fig. 5a)—providing actionable insights into individual cardiorespiratory performance. By capturing and analyzing these multimodal signals, the SCP system offers a detailed picture of how the body adapts to physical activity, laying the foundation for personalized exercise planning and performance optimization.

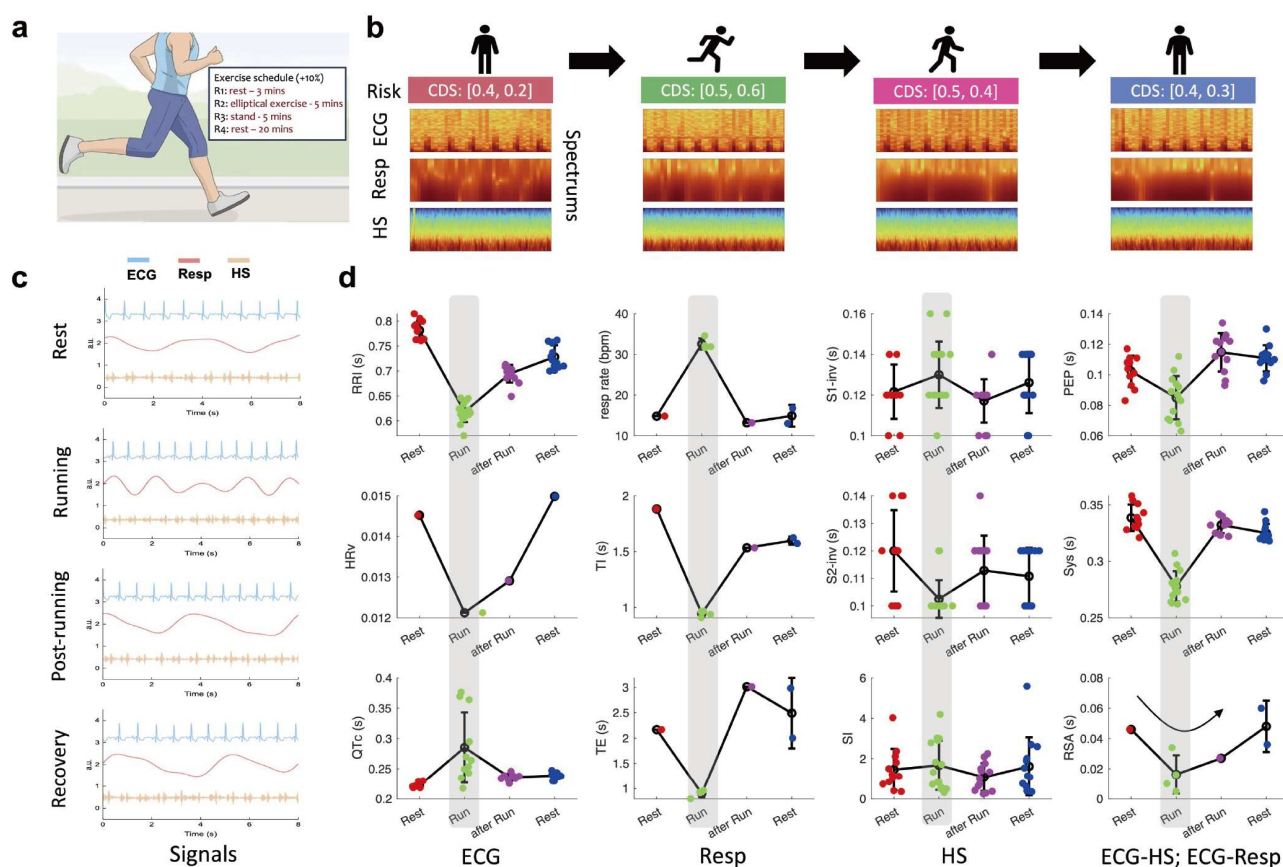


Figure 5 Cardiopulmonary function monitoring during exercise optimization. (a) A schematic illustrating daily exercise scenarios incorporating our SCP system. (b) Integration of the ECG, HS, and respiratory spectra to monitor dynamic physiological states in a single individual across exercise phases (pre-exercise (rest), exercise, post-exercise, and recovery); CDS is the cardiopulmonary dysfunction score of the cardiac and respiratory risk probability generated from the MCMF model. (c) Synchronized ECG, HS, and respiratory signals during different exercise phases. (d) 12 cardiopulmonary parameters were extracted from the multimodal signals during the progress of the exercise.

Spectral and temporal analyses revealed distinct cardiorespiratory dynamics (Fig. 5b, c). In this experiment, similar to the device signal validation experiment, the SCP system was positioned at the precordial site, capturing high-fidelity multimodal data from ECG, HS, and respiratory signals. 12 key parameters are extracted, enabling precise assessment of cardiorespiratory function for personalized exercise planning and risk prediction. As shown in Fig. 5d, during exercise, HR (i.e., 60/RRI) increased significantly (mean increase of 21 bpm, $P < 0.01$), with a corresponding reduction in RRI and prolongation of the QTc interval ($\Delta\text{QTc} = +67$ ms), reflecting an extended cardiac ejection cycle to meet elevated oxygen demands. Concurrently, RR surged (mean 32 bpm, $P < 0.01$), with synchronized shortening of inhalation and exhalation durations (I:E ratio approaching 1:1), indicative of enhanced ventilatory efficiency. HS analysis showed a prolonged S1 interval ($\Delta\text{S1} = +10$ ms) and reduced S2 interval ($\Delta\text{S2} = -15$ ms), suggesting increased mechanical stress on cardiac valves to sustain elevated cardiac output. Cardiorespiratory interactions, quantified via PEP and systolic time (Sys), decreased during exercise ($\Delta\text{PEP} = -20$ ms, $\Delta\text{Sys} = -64$ ms), indicating accelerated pulse wave velocity and transient arterial stiffening. RSA amplitude declined ($\Delta\text{RSA} = -65\%$, $P < 0.05$), reflecting tighter heart-lung coupling to maintain autonomic balance under physiological stress.

In the recovery phase, HR and RR gradually normalized, with PEP rebounding ($\Delta\text{PEP} = +6$ ms), signaling restored vascular elasticity. The PEP/left ventricular ejection time (LVET) ratio peaked post-exercise (>0.4 , $P < 0.01$), suggesting transient cardiac strain, but normalized during recovery, indicating robust cardiac recovery. RSA values approached resting levels (within 10% of baseline), reflecting restored autonomic regulation.

Comparative analyses across exercise states, powered by the machine learning model, reveal the SCP system's ability to detect subtle yet significant physiological changes (Fig. 5b). Unlike conventional wearables that rely on single-modal metrics such as HR or RR, our precordial SCP system delivers a holistic view of cardiorespiratory function. By quantifying key parameters like HR, respiratory efficiency, and heart-lung coupling, this platform empowers individuals to evaluate the effectiveness of their exercise routines and optimize training procedures. With its ability to provide real-time feedback and identify risks of over-exertion, the SCP system represents a transformative step toward precision health monitoring and enhanced cardiorespiratory fitness.

CONCLUSIONS

Conventional wearable devices for cardiopulmonary assessment are limited by challenges in synchronously integrating multi-

modal sensors, robustly fusing heterogeneous signal streams, and accurately detecting anomalies in cardiopulmonary interactions amidst inherent physiological variability. To address these gaps, we developed a compact, lightweight SCP system that combines high-sensitivity multimodal sensing with a sophisticated multi-criteria fusion model. By synchronously capturing ECG, HS, and respiratory signals, the SCP system delivers over 12 critical cardiopulmonary parameters with robust anomaly detection, overcoming the constraints of traditional peripheral wearables.

The SCP system features a flexible, high-sensitivity multimodal sensing layer that precisely captures ECG, HS, and respiratory data, seamlessly integrated with a custom-engineered circuit layer for efficient data acquisition, signal processing, and wireless transmission. Beyond its hardware innovations, the system leverages a novel physiological modeling framework that exploits the interplay among ECG, HS, and respiratory signals to extract comprehensive cardiopulmonary metrics and detect dysfunction. By constructing a multi-modal, multi-criteria fusion model and integrating diverse, publicly available datasets across independent modalities, we achieved a high accuracy for detecting abnormal cardiopulmonary function. Moreover, we validated the system's utility in real-time exercise monitoring, capturing dynamic physiological changes across pre-exercise, exercise, post-exercise, and recovery phases. Future investigations will assess signal interference during high-intensity exercise and long-term stability tests, thereby enhancing its practical applicability. The platform provides critical insights into dynamic physiological responses, offering a valuable reference for optimizing exercise regimens, a powerful tool for optimizing exercise routines, identifying early signs of dysfunction, and broadening the application scope of wearable biomedical technologies across diverse healthcare and fitness domains.

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Conflict of interest The authors declare that they have no conflict of interest.

Supplementary information Experimental details and supporting data are available in the online version of the paper.

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一种用于实时心肺监测和异常检测的多模态智能胸贴

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摘要 心肺稳态对健康至关重要, 其紊乱常提示早期病理变化。然而, 当前可穿戴设备因聚焦于单一或少数模态, 难以全面捕捉这些关键交互的完整谱系。本研究提出一种多模态智能胸贴(SCP)系统, 将柔性传感模块与多准则多模态融合(MCMF)机器学习模型相结合。该贴片(重量5.4 g, 厚度3.6 mm)可同时监测心电图、心音和呼吸信号, 实现12项心肺参数的实时提取。与商用设备相比, 该贴片能保持稳定信号质量, 并用于不同个体的参数测量。我们在包含475名参与者共5,561条记录的扩展数据集上验证MCMF在心肺异常检测的性能, 实现87%识别准确率, 超越传统方法。最后, 我们将SCP用于实时运动监测, 揭示了动态生理变化($\Delta\text{HR} = 21$ 次/min, $\Delta\text{PEP} = -30$ ms), 且信号保真度优异, 因此在可扩展个性化健康管理领域具有重要应用潜力。